The electromagnetic spectrum

3a

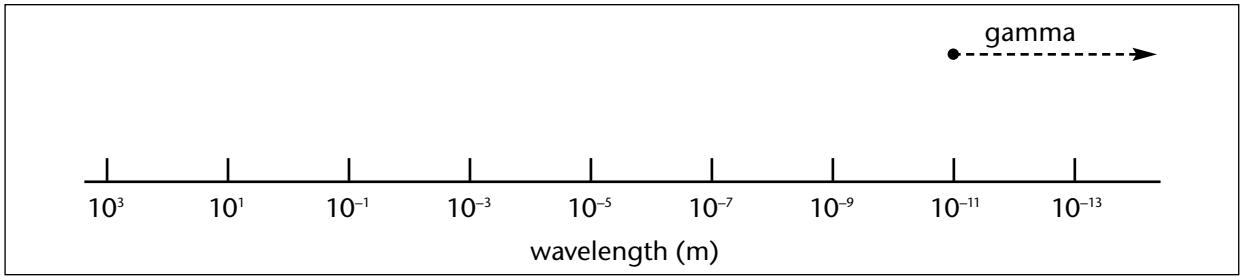
Gamma rays, infra-red waves, light rays, microwaves, radio waves, ultra-violet rays and X-rays are similar. They all travel at the speed of light (300 000 000 m/s or 3×10^8 ms^{-1}) and they all have an electric and a magnetic wave component. However, they are different because they vary in wavelength, frequency and energy. They are also produced by different processes and have different effects on us and other things.

1 Here is the list again. This time the radiation list shows the wavelength of each radiation type: gamma rays ($< 1 \times 10^{-11}$ m), infra-red waves ($7 \times 10^{-7} - 1 \times 10^{-3}$ m), light ($4 \times 10^{-7} - 7 \times 10^{-7}$ m), microwaves ($1 \times 10^{-3} - 1 \times 10^{-1}$ m), radio waves ($> 1 \times 10^{-1}$ m), ultra-violet rays ($1 \times 10^{-8} - 4 \times 10^{-7}$ m) and X-rays ($1 \times 10^{-11} - 1 \times 10^{-8}$ m). Rewrite the list. Begin with the longest wavelength.

Longest _____

_____ shortest

2 The diagram below shows a wavelength scale. Draw short, labelled lines like that shown for the gamma ray to indicate the wavelength range of the other electromagnetic (EM) waves.



3 Wave frequency is measured in hertz. A frequency of one hertz (1 Hz) means one cycle of the wave in one second. The list of electromagnetic waves is given again. This time their frequency ranges are shown: gamma rays ($> 3 \times 10^{19}$ Hz), infra-red waves ($3 \times 10^{11} - 4 \times 10^{14}$ Hz), light ($4 \times 10^{14} - 7 \times 10^{14}$ Hz), microwaves ($3 \times 10^{9} - 3 \times 10^{11}$ Hz), radio waves ($< 3 \times 10^{9}$ Hz), ultra-violet rays ($7 \times 10^{14} - 3 \times 10^{16}$ Hz) and X-rays ($3 \times 10^{16} - 3 \times 10^{19}$ Hz). Rewrite the list beginning with the smallest frequency.

Smallest _____

_____ highest

4 Compare the two lists. What do you notice?

_____

5 One joule (1 J) is a unit of energy equal to the work done when a force of one newton (1 N) acts through a distance of one metre (1 m). It's the work you do when you lift a 100 g bar of chocolate from your trouser pocket to your mouth. The energy ranges of the different EM radiations are listed after each radiation type: gamma rays ($> 2 \times 10^{-14}$ J), infra-red waves ($2 \times 10^{-22} - 3 \times 10^{-19}$ J), light ($3 \times 10^{-19} - 5 \times 10^{-19}$ J), microwaves ($2 \times 10^{-24} - 2 \times 10^{-22}$ J), radio waves ($< 2 \times 10^{-24}$ J), ultra-violet rays ($5 \times 10^{-19} - 2 \times 10^{-17}$ J) and X-rays ($2 \times 10^{-17} - 2 \times 10^{-14}$ J). Rewrite the list, ordering it by energy level.

Smallest _____

_____ highest

6 How does this list compare with your other lists?

_____

7 Consider this object list: atom, bee, house, molecule, nucleus (of an atom), pinhead, plant cell. Match this list with the ordered wavelength list of EM radiation. (Question 1.)

Radiation list: _____

Object list: _____

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The electromagnetic spectrum

3b

8 The following list states sources of the various types of EM radiation: electric radiator, light bulb, microwave oven, nuclear reactor, quartz lamp, radio transmitter, X-ray machine. Match this list with the ordered wavelength list of EM radiation.

Radiation list: _____

Source list: _____

9 The following list indicates possible outcomes or uses of the various types of EM radiation: communication, cooking, death, deformities, heat, skin cancer, sight. Match this list with the ordered wavelength list of EM radiation.

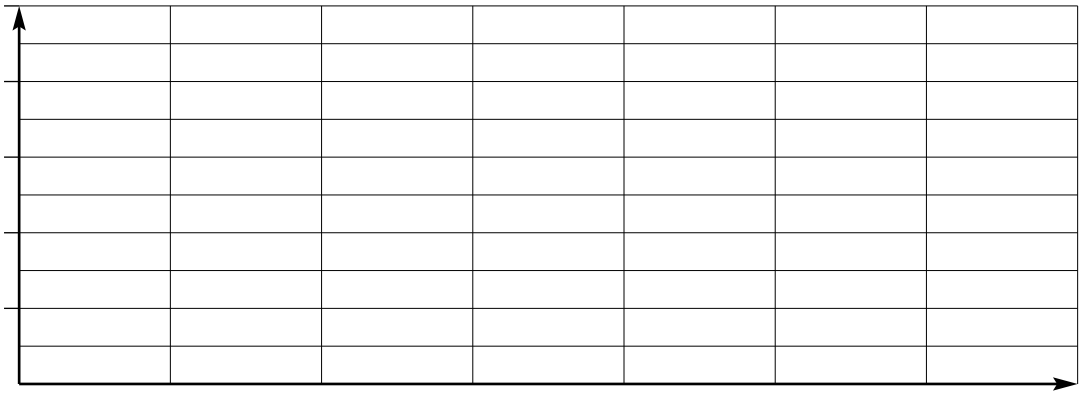
Radiation list: _____

Outcome list: _____

10 Much solar radiation is trapped by the earth's atmosphere. The table records the average atmospheric heights at which various types of electromagnetic radiation are mostly absorbed.

Radiation type	**Gamma**	**Infra-red**	**Light**	**Microwave**	**Radio** waves	X-rays	Ultra-violet
Absorption altitude (km)	25	12	0	25	0	100	30

Show this information in a column graph. List the radiation types in order of wavelength on the x-axis.



Note: Some ultra-violet rays as well as some microwaves do reach the surface of the earth.

11 Cross out the incorrect term from each set of brackets.

The ozone layer in the (atmosfere / atmosphere) is a thin, low altitude (laier / layer) of ozone gas, O_3, a form of oxygen. It is able to (absorb / absorp) ultra-violet rays. However, the (ozone / ozoone) layer has thinned out (during / while) the last few (decades / decads). As a result during spring a large ozone hole (developes / develops) above Antarctica and a smaller (one / own) above the Arctic. These holes (rebuild / rebuilt) again over summer.

(Unfortunately / Unfortunatly) the ozone layer is also somewhat (fin / thin) above southern parts of (Australia / Austria), especially during spring and (earlie / early) summer. (Sunscrean / Sunscreen), hats, sunglasses and long-sleeved (clotheing / clothing) offer much (protection / protektion) against these (dangerous / harmless) ultra-violet rays. Chlorofluorocarbons and halons are (beleaved / believed) to have caused the (los / loss) of ozone. Cold temperatures (increase / increese) the speed of ozone (depleetion / depletion). If the (hole / whole) above the Arctic were to (increase / increese), large (people / populations) in northern Europe, northern (asia / Asia) and northern North America (could / would) be threatened (by / with) melanomas caused by (ultra-violet / ultrared) radiation.

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